

CS/B.Tech/IT/Even/Sem-6th/IT-603/2015



WEST BENGAL UNIVERSITY OF TECHNOLOGY

IT-603

SOFTWARE ENGINEERING

Time Allotted: 3 Hours

Full Marks: 70

The questions are of equal value.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All symbols are of usual significance.

GROUP A

(Multiple Choice Type Questions)

1. Answer any *ten* questions. 10 × 1 = 10
- (i) Waterfall model is not suitable for
☒ (A) Accommodating change ☐ (B) Complex Projects
☐ (C) Small projects ☐ (D) None of above
- (ii) Which of the following life cycle model deals with risk associated with software products?
☐ (A) Prototype model ☒ (B) Spiral model
☒ (C) Incremental model ☐ (D) Waterfall model
- (iii) Coding and testing are done in following manner
☒ (A) Ad-hoc ☐ (B) Cross sectional
☐ (C) Bottom up ☐ (D) Top-down
- (iv) Which of the following is not the characteristic of software?
☐ (A) Software does not wear out ☒ (B) Software is always correct
☒ (C) Software is flexible ☐ (D) Software is not manufactured

6304

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- (v) A Context diagram
☐ (A) describes the context of a system
☐ (B) is a DFD that gives an overview of a system
☐ (C) is a detailed description of a system
☐ (D) is the base of object oriented design
- (vi) System testing performed by a set of friendly customers is called
☐ (A) Usability testing ☐ (B) Alpha testing
☐ (C) Smoke testing ☒ (D) Beta testing
- (vii) COCOMO refers to
☐ (A) Cost Construction Model
☒ (B) Constructive Cost Model
☐ (C) Corporate Construction Model
☐ (D) Corporate Cost Model
- (viii) Structured English is
☐ (A) structured programming language
☐ (B) description of process in simple English
☐ (C) natural language based algorithmic language
☐ (D) structured query language
- (ix) If data from one module is used to direct the order of execution in another, then the coupling is known as
☐ (A) Stamp Coupling ☐ (B) Data Coupling
☒ (C) Control Coupling ☐ (D) Content Coupling
- (x) Critical path method is a task of
☐ (A) Project Planning ☒ (B) Project Scheduling
☐ (C) Risk management ☐ (D) Configuration management
- (xi) What is the most expensive phase in a software development life cycle?
☐ (A) Design ☒ (B) Maintenance
☐ (C) Analysis ☐ (D) Testing
- (xii) A Class consists of which of these abstractions?
☐ (A) Set of the objects ☐ (B) Operations
☐ (C) Attributes ☒ (D) All of the above

6304

2

CS/B.Tech/IT/Even/Sem-6th/IT-603/2015

GROUP B
(Short Answer Type Questions)

Answer any *three* questions.

2. What are the different levels of testing and their goals? $3 \times 5 = 15$
3. What is feasibility study? Explain different types of feasibility study. 5
4. What is Use-case diagram? Draw the Use-case diagram of 'Railway reservation system'. $2+3$
5. Why COCOMO is called Heuristic Estimation Technique? Assume that the size of an organic type software product has been estimated to be 48000 lines of source codes. Assume that the average salary of software engineers is Rs. 18,000 per month. Determine the effort required to develop the software product; total cost and the nominal develop time. 5
6. Explain the phases of Classical Waterfall Model with advantages and disadvantages. $1+4$
- 3+1+1

GROUP C
(Long Answer Type Questions)

Answer any *three* questions.

- (a) Draw the control flow graph for the following program. Also calculate the cyclomatic complexity. $3 \times 15 = 45$
- ```

int gcd(int x, int y)
{
 while(x != y)
 {
 if(x > y)
 x = x - y;
 else
 y = y - x;
 }
 return x;
}

```
- (b) What do you mean by Activity Diagram? Explain with an example. Why is it used?  $5$
- (c) Distinguish between verification and validation.  $5$

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6304

3

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8. (a) Draw the context diagram and Level-1 DFD for Online Banking Management System. Draw also Use-Case diagram for this system.  $5+2$
- (b) What is SDLC? Explain all phases.  $5$
- (c) Briefly explain team structures and organization process.  $3$
9. (a) What do you mean by requirement analysis?  $3+5+5+2$
- (b) Explain the phases of Spiral Model with advantages and disadvantages.
- (c) Discuss the different stages of 'Capability Maturity Model'.
- (d) What do you mean by Software Products?
10. (a) What is estimation? Explain its role in S/W planning. Explain any one cost estimation model with its merits and demerits.  $8$
- (b) Differentiate between logical DFD and physical DFD.  $3$
- (c) State true or false with reason:  $4$
- (i) S/W is developed or engineered; it is not manufactured in the classical sense.
- (ii) S/W does not wear out.
11. Write short notes on any *three* of the following:  $3 \times 5$
- (a) Prototyping Model
- (b) Stubs and Drivers in Integration Testing
- (c) Software Quality Assurance
- (d) Project Staffing
- (e) Black Box Testing.

6304

4